

PRACTICAL-3: Mann-Whitney U Test

1 Objective

To compare the effectiveness of two independent teaching methods on students' reading comprehension scores using the Mann-Whitney U test.

2 Problem Statement

A study was conducted to determine whether there is a difference in reading comprehension performance among 4th-grade students taught using two different instructional methods after four weeks of intervention:

- **Method 1:** Pull-out program (30 minutes per day, 4 days a week)
- **Method 2:** Small group classroom instruction (45 minutes per day, 4 days a week)

The reading comprehension scores obtained from the two independent groups are not assumed to be normally distributed, and the sample sizes are small. Therefore, a non-parametric test is required to assess whether a significant difference exists.

Given Data:

- **Method 1 (X):** 48, 40, 39, 50, 41, 38, 53
- **Method 2 (Y):** 14, 18, 20, 10, 12, 102, 17

3 Theory: Mann-Whitney U Test

3.1 Introduction to the Test

Developed by H.B. Mann and D.R. Whitney in 1947, this test is a non-parametric alternative to the usual two-sample independent t-test. It is used to compare two independent groups when the data is non-normal or the sample size is small.

3.2 Assumptions

To apply this test, the following conditions must be met:

- The two samples being compared must be independent of each other.
- The variable under consideration should have a continuous distribution.

3.3 Hypotheses

Let there be two independent random samples, $X_1, X_2, \dots, X_m$ (of size m) and $Y_1, Y_2, \dots, Y_n$ (of size n), drawn from two populations with distribution functions $F(x)$ and $G(y)$, respectively.

The fundamental null hypothesis is that the two population distributions are identical (i.e., the two treatment effects do not differ significantly).

- **Null Hypothesis (H_0):** $F(x) = G(y)$ (There is no significant difference in scores between Method 1 and Method 2).
- **Alternative Hypothesis (H_1):** $F(x) \neq G(y)$ (There is a significant difference in scores between Method 1 and Method 2).
Note: One-tailed alternatives would be $F(x) > G(y)$ or $F(x) < G(y)$.

3.4 Procedure & Test Statistic

1. **Combine & Rank:** Combine both samples into a single sequence of $(m+n)$ observations. Arrange them in increasing order of magnitude and assign ranks from 1 to $(m+n)$.
2. **Sum the Ranks:** Separate the ranks back into their original groups.
 - Let $R_x = \sum r_x$ be the sum of the ranks for the X observations.
 - Let $R_y = \sum r_y$ be the sum of the ranks for the Y observations.
3. **Calculate U Values:** Compute the intermediate statistics U_x and U_y :

$$U_x = mn + \frac{m(m+1)}{2} - R_x$$

$$U_y = mn + \frac{n(n+1)}{2} - R_y$$

4. **Test Statistic (U):** The final Mann-Whitney test statistic is defined as the minimum of the two values:

$$U = \min(U_x, U_y)$$

3.5 Decision Rule

- **Manual Approach:** Reject H_0 if $U \leq U_{\alpha, m, n}$, where $U_{\alpha, m, n}$ is the critical value obtained from the Mann-Whitney U-Test tables for a given level of significance α .
- **Software Approach:** Reject H_0 if the calculated p-value $< \alpha$.

3.6 Important Mathematical Remarks

- In every test, the sum of the U statistics equals the product of the sample sizes: $U_x + U_y = mn$.
- The value of U is strictly bounded: $0 \leq U \leq mn$.
- $U = 0$ when all X_i values are larger than all Y_j values.
- $U = mn$ when all X_i values are smaller than all Y_j values.
- If H_0 is true (populations are equal), we expect the rank sums R_x and R_y to be similar relative to their sample sizes.

4 R Implementation & Calculation

We execute the test in R using the `wilcox.test` function, specifying that the two samples are independent (which is the default when `paired = FALSE`).

6 Conclusion

Based on the statistical analysis:

- **Test Result:** The calculated test statistic is $W = 42$ with a p-value = 0.026.
- **Decision:** Since the p-value (0.026) is strictly less than our standard level of significance ($\alpha = 0.05$), the null hypothesis (H_0) is rejected.
- **Inference:** There is a significant difference between the two teaching methods. The teaching methods do not produce equal reading comprehension outcomes.

PRACTICAL-4: KS Test

Aim:

To test statistical hypotheses concerning continuous distributions using the Kolmogorov-Smirnov (K-S) one-sample and two-sample tests.

PART I: Kolmogorov-Smirnov One-Sample Test

7 Problem Statement

20 observations were chosen randomly from the continuous uniform distribution over $(0, 1)$, recorded to four significant figures, and rearranged in increasing order of magnitude. Determine the value of the Kolmogorov-Smirnov statistic D_n and test the null hypothesis that the square roots of these numbers also have the continuous uniform distribution over $(0, 1)$ at a 5% level of significance.

Given Data (x):

0.0123, 0.1039, 0.1954, 0.2621, 0.2802, 0.3217, 0.3645, 0.3919, 0.4240, 0.4814, 0.5139, 0.5846, 0.6275, 0.6541, 0.6889, 0.7621, 0.8320, 0.8871, 0.9249, 0.9634

Transformation: Let $Y = \sqrt{X}$

8 Theory & Mathematical Foundation

Introduction to the Test:

The Kolmogorov-Smirnov one-sample test (developed by Andrey Kolmogorov and Nikolai Smirnov) is a goodness-of-fit procedure. It compares the empirical (observed) cumulative distribution function (CDF) of a sample with a completely specified theoretical continuous distribution, $F_0(x)$ (such as normal, uniform, or exponential).

Use Cases:

- To check goodness-of-fit for a specified continuous distribution.
- To validate distributional assumptions.
- It is particularly suitable when no parameters are estimated from the data.

Procedure & Formulas:

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a population having distribution function $F(x)$. Let $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ denote the corresponding order statistics.

Step 1: Define the Empirical Distribution Function, $F_n(x)$

$F_n(x)$ is a step function representing the proportion of sample observations less than or equal to x :

$$F_n(x) = \begin{cases} 0 & \text{for } x < X_{(1)} \\ \frac{k}{n} & \text{for } X_{(k)} \leq x < X_{(k+1)}, \text{ where } k = 1, 2, \dots, n-1 \\ 1 & \text{for } x \geq X_{(n)} \end{cases}$$

Step 2: Set Up the Hypotheses

- **Null Hypothesis (H_0):** $F(x) = F_0(x) \quad \forall x$ (The data follows the specified distribution).
- **Alternative Hypothesis (H_1):** $F(x) \neq F_0(x)$ for some x (Two-sided test).

Step 3: Define the Test Statistic

The appropriate test statistic for the two-tailed hypothesis is D_n , which represents the maximum vertical deviation between the empirical CDF and the theoretical CDF.

$$D_n = \max_x |F_n(x) - F_0(x)|$$

To simplify computation from the sample order statistics, we compute two one-sided statistics, D_n^+ and D_n^- :

$$D_n^+ = \max_{1 \leq i \leq n} \left[\frac{i}{n} - F_0(X_{(i)}) \right]$$
$$D_n^- = \max_{1 \leq i \leq n} \left[F_0(X_{(i)}) - \frac{i-1}{n} \right]$$

The two-sided test statistic is simply the maximum of these two values:

$$D_n = \max(D_n^+, D_n^-)$$

If H_0 is true, D_n should be small. A large value of D_n indicates a significant discrepancy between the observed and expected distributions.

Step 4: Decision Rule

Reject H_0 if $D_n > C$, where C is the critical value from the K-S tables at a given level of significance α . Alternatively, using software, reject H_0 if the calculated p-value $< \alpha$.

9 R Implementation & Calculation (One-Sample)

R Code:

```
# Input the given observations
x <- c(0.0123, 0.1039, 0.1954, 0.2621, 0.2802, 0.3217, 0.3645, 0.3919, 0.4240,
      0.4814, 0.5139, 0.5846, 0.6275, 0.6541, 0.6889, 0.7621, 0.8320, 0.8871,
      0.9249, 0.9634)

# Calculate the square roots of the observations
y <- sqrt(x)

# Perform the K-S one-sample test against the Uniform(0,1) distribution
ks.test(y, "punif", 0, 1)
```

Result Output:

```
Exact one-sample Kolmogorov-Smirnov test

data:  y
D = 0.36196, p-value = 0.007456
alternative hypothesis: two-sided
```

Conclusion (Experiment 1):

The calculated value of the test statistic is $D_n = 0.36196$, and the corresponding p-value is 0.007456. Since the p-value (0.007456) is strictly less than the 5% level of significance, we reject the null hypothesis. We conclude at the 5% level of significance that the square roots of the observations do not follow a Uniform $(0, 1)$ distribution.

PART II: Kolmogorov-Smirnov Two-Sample Test

1 Problem Statement

It is known that the distribution of a standardized chi-square variable with large degrees of freedom can be approximated by the standard normal distribution. This experiment investigates the agreement between these two distributions for moderate degrees of freedom ($v = 18$).

Two mutually independent random samples, each of size eight, were generated.

Given Data:

- **Sample 1 (Standard Normal):** -1.91, -1.22, -0.96, -0.72, 0.14, 0.82, 1.45, 1.86
- **Sample 2 (Chi-square, $v = 18$):** 4.90, 7.25, 8.04, 14.10, 18.30, 21.21, 23.10, 28.12

Transformation needed: Standardize the Chi-square values using

$$Z = \frac{X - v}{\sqrt{2v}}$$

2 Theory: K-S Two-Sample Test

The Kolmogorov-Smirnov two-sample test is used to determine whether two independent samples are drawn from the same continuous underlying distribution.

Use Cases:

Comparing two empirical distributions, validating distributional approximations, suitable for small sample sizes, and requires no assumption of normality.

Hypotheses:

- H_0 : Both samples come from the same distribution ($F_1(x) = F_2(x)$).
- H_1 : The two samples come from different distributions ($F_1(x) \neq F_2(x)$).

3 R Implementation & Calculation (Two-Sample)

R Code:

```

# Input the two independent samples
normal <- c(-1.91, -1.22, -0.96, -0.72, 0.14, 0.82, 1.45, 1.86)
chi_sq <- c(4.90, 7.25, 8.04, 14.10, 18.30, 21.21, 23.10, 28.12)

# Standardize the chi-square variables (degrees of freedom v = 18)
v <- 18
chi_std <- (chi_sq - v) / sqrt(2 * v)

# Perform the K-S two-sample test
ks.test(normal, chi_std)

```

Result Output:

Exact two-sample Kolmogorov-Smirnov test

```

data: normal and chi_std
D = 0.25, p-value = 0.9801
alternative hypothesis: two-sided

```

Conclusion (Experiment 2):

The calculated test statistic is $D = 0.25$, and the p-value is 0.9801. Because the p-value (0.9801) is much greater than the level of significance (0.05), we accept the null hypothesis. We conclude that the standardized chi-square distribution with moderate degrees of freedom ($v = 18$) shows reasonable agreement with the standard normal distribution, thus supporting the normal approximation.

PRACTICAL 5: Kruskal-Wallis Test

1 Aim

To test the hypothesis concerning the central tendencies of multiple independent groups using the Kruskal-Wallis test.

2 Problem Statement (Experiment)

Three independent samples were taken from three populations to study whether there is a significant difference among their central tendencies at a 5% level of significance ($\alpha = 0.05$).

Given Data:

- **Group A:** 12, 15, 14, 13
- **Group B:** 18, 20, 17, 19
- **Group C:** 25, 22, 24, 23

3 Theory & Mathematical Foundation

3.1 Introduction to the Test

To test whether several independent samples come from identical populations, the standard parametric procedure is the One-Way ANOVA, provided that assumptions of population normality and homogeneity of variances are fulfilled. If these assumptions are violated, the Kruskal-Wallis test (Kruskal & Wallis, 1952) is utilized.

It is a rank-based, non-parametric alternative to the one-way ANOVA. It can be viewed as an extension of the Mann-Whitney U-test (which is restricted to only two independent samples) designed to handle one-way classification data for three or more independent groups.

3.2 Use Cases

- When comparing three or more independent groups on a numeric outcome.
- When sample sizes are small or unequal.
- For non-normal or skewed data (or when the assumption of homogeneity of variances is strictly violated).
- For ordinal variables.

3.3 Hypotheses

Let $\eta_1, \eta_2, \dots, \eta_k$ represent the medians of the k populations.

- **Null Hypothesis (H_0):** All groups come from the same population (their distributions are identical).

$$H_0 : \eta_1 = \eta_2 = \dots = \eta_k$$

- **Alternative Hypothesis (H_1):** At least one group comes from a different population (at least one distribution differs).

$$H_1 : \eta_i \neq \eta_j \quad \text{for some } i \neq j$$

3.4 Procedure & Test Statistic

Let $n_1, n_2, \dots, n_k$ denote the sample sizes selected from k populations, and let $N = n_1 + n_2 + \dots + n_k$ denote the total number of observations.

1. Combine the samples of N observations and arrange them in increasing order of magnitude.
2. Assign ranks from 1 to N . The sum of all ranks will naturally be $\frac{N(N+1)}{2}$.
3. Let $R_1, R_2, \dots, R_k$ be the sum of the ranks for the observations belonging to the $1^{st}, 2^{nd}, \dots, k^{th}$ samples respectively.

The Kruskal-Wallis test statistic, H , is calculated using the following formula:

$$H = \frac{12}{N(N+1)} \left[\frac{R_1^2}{n_1} + \frac{R_2^2}{n_2} + \dots + \frac{R_k^2}{n_k} \right] - 3(N+1)$$

Alternatively written as:

$$H = \frac{12}{N(N+1)} \left[\sum_{i=1}^k \frac{R_i^2}{n_i} \right] - 3(N+1)$$

3.5 Distribution & Decision Rule

Under the null hypothesis H_0 , for sufficiently large sample sizes (typically $n_i \geq 5$), the test statistic H follows approximately a Chi-square distribution with $k - 1$ degrees of freedom:

$$H \sim \chi_{k-1}^2$$

Decision Rule: Reject H_0 if the calculated statistic $H \geq \chi_{(k-1), \alpha}^2$ at the predetermined α -level of significance, or if the computational p-value $< \alpha$.

4 R Implementation & Calculation

To perform the test computationally, we define the data points for each group, combine them into a single vector, and create a corresponding factor variable to identify group membership.

R Code:

```

# Input the sample data
A <- c(12, 15, 14, 13)
B <- c(18, 20, 17, 19)
C <- c(25, 22, 24, 23)

# Combine the data and create grouping factors
data <- c(A, B, C)
group <- factor(rep(c("A", "B", "C"), each = 4))

# Perform the Kruskal-Wallis Rank Sum Test
kruskal.test(data ~ group)

```

5 Result & Output

Executing the code in the R console yields the following output:

Result Output:

```

      Kruskal-Wallis rank sum test

data:  data by group
Kruskal-Wallis chi-squared = 9.8462, df = 2, p-value = 0.007277

```

6 Interpretation & Conclusion

- **Test Statistic:** The calculated value of the test statistic (H or chi-squared) is 9.8462 with 2 degrees of freedom ($k - 1 = 3 - 1 = 2$).
- **Decision:** We compare the calculated p-value to our level of significance ($\alpha = 0.05$). The calculated p-value is 0.007277.
- **Conclusion:** Since the p-value (0.007277) is strictly less than the level of significance (0.05), we reject the null hypothesis (H_0). We conclude at the 5% level of significance that there is a significant difference among the central tendencies of the groups; at least one group comes from a different population.

PRACTICAL-6: Kendall's Rank Correlation (τ)

1 Aim

To test the hypothesis regarding the association between two ordinal variables using Kendall's Tau (τ).

2 Problem Statement (Experiment)

Two judges ranked nine teas on the basis of taste and full-bodied properties, with (1) indicating the highest ranking. Calculate the Kendall measure of association, test the null hypothesis of independence, and find the appropriate one-tailed p-value.

Given Data:

Tea	A	B	C	D	E	F	G	H	I
Judge 1	1	5	9	7	4	6	8	2	3
Judge 2	4	3	6	8	2	7	9	1	5

3 Theory & Mathematical Foundation

3.1 Introduction to the Test

Kendall's Tau (τ) is a non-parametric measure of rank correlation that evaluates the strength and direction of monotonic association between two ordinal variables. Unlike Spearman's correlation (which relies on the deviations of ranks), Kendall's Tau is based on comparing all possible pairs of observations and classifying them as either concordant or discordant.

3.2 Concordant and Discordant Pairs

For any pair of observations (x_i, y_i) and (x_j, y_j) :

- **Concordant Pairs (C):** A pair is concordant if the ranks for both elements agree in their directional sort. That is, if both $x_i > x_j$ and $y_i > y_j$, or if both $x_i < x_j$ and $y_i < y_j$. (Ranked in the same order).
- **Discordant Pairs (D):** A pair is discordant if the ranks for both elements disagree. That is, if $x_i > x_j$ but $y_i < y_j$, or vice versa. (Ranked in the opposite order).

The Kendall Tau coefficient is mathematically defined as:

$$\tau = \frac{C - D}{\frac{1}{2}n(n-1)}$$

Where n is the sample size, and $\frac{1}{2}n(n-1)$ represents the total number of possible pairs.

3.3 Hypotheses

- **Null Hypothesis (H_0):** The ranking of teas by the two judges is independent.

$$H_0 : \tau = 0$$

- **Alternative Hypothesis (H_1):** The ranking of teas by the two judges is positively associated (one-tailed test).

$$H_1 : \tau > 0$$

3.4 Level of Significance

$$\alpha = 0.05 \text{ (5\%)}$$

4 R Implementation & Calculation

To perform this computationally, we input the ranks provided by the two judges and use the `cor.test` function, specifying the method as "kendall" and the alternative hypothesis as "greater" to match our one-tailed assumption.

R Code:

```
# Input the rank data for both judges
judge_1 <- c(1, 5, 9, 7, 4, 6, 8, 2, 3)
judge_2 <- c(4, 3, 6, 8, 2, 7, 9, 1, 5)

# Perform Kendall's rank correlation test (one-tailed: positive association)
result <- cor.test(judge_1, judge_2, method="kendall", alternative="greater")

# Print the result
print(result)
```

5 Result & Output

Executing the R code yields the following output:

Result Output:

```
Kendall's rank correlation tau

data:  judge_1 and judge_2
T = 28, p-value = 0.02231
alternative hypothesis: true tau is greater than 0
sample estimates:
      tau
0.5555556
```

(Note for your exam: The test statistic $T = 28$ given by R in this specific output represents the number of strictly Concordant pairs (C).)

6 Interpretation & Conclusion

- **Test Statistics:** The value of the test statistic is $T = 28$, and the sample estimate for Kendall's Tau is $\tau \approx 0.5556$.
- **Decision Rule:** We compare the calculated p-value to our level of significance ($\alpha = 0.05$). The calculated p-value is 0.02231.
- **Conclusion:** Since the p-value (0.02231) is strictly less than 0.05, we reject the null hypothesis (H_0) at the 5% level of significance. We conclude that the rankings of the teas by the two judges are positively associated (i.e., $\tau > 0$).

PRACTICAL-7: Spearman's Rank Correlation Coefficient

1 Aim

To apply Spearman's Coefficient of Rank Correlation to evaluate the relationship between two variables.

2 Problem Statement (Experiment)

Company-financed expenditures in manufacturing on research and development (R&D) are currently about 2.7% of sales in Japan and 2.8% of sales in the United States. However, when these figures are looked at separately according to industry, the data (from Mansfield, 1989) show some large differences.

Determine whether there is a significant positive relationship between the percentages spent by Japan and the United States using Spearman's coefficient of rank correlation.

Given Data (R&D as % of Sales):

Industry	Japan	United States
Food	0.8	0.4
Textiles	1.2	0.5
Paper	0.7	1.3
Chemicals	3.8	4.7
Petroleum	0.4	0.7
Rubber	2.9	2.2
Ferrous metals	1.9	0.5
Non-ferrous metals	1.9	1.4
Metal products	1.6	1.3
Machinery	2.7	5.8
Electrical equipment	5.1	4.7
Motor vehicles	3.0	3.2
Other transport equipment	2.6	1.2
Instruments	4.5	9.0

3 Theory & Mathematical Foundation

3.1 Introduction to the Test

Spearman's rank correlation coefficient (ρ or r_s) is a non-parametric measure of statistical dependence or association between two variables. It measures the strength and direction of a monotonic relationship based on the ranked values of the data rather than the raw data itself.

3.2 Key Points:

- **Use Cases:** Used when data are not normally distributed, when the data is ordinal, or when dealing with non-linear but monotonic relationships.

- **Rank-Based:** It relies entirely on the ranked positions of the values within their respective datasets.
- **No Linearity Required:** Unlike Pearson's correlation, Spearman's does not require the relationship to be strictly linear.
- **Range:** The calculated value always lies between -1 and $+1$, where $+1$ indicates a perfect positive monotonic relationship and -1 indicates a perfect negative monotonic relationship.

3.3 Mathematical Formula

When there are no tied ranks, Spearman's correlation coefficient is calculated as:

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where d_i is the difference between the two ranks of each observation, and n is the number of observations.

3.4 Hypotheses & Level of Significance

- **Null Hypothesis (H_0):** There is no significant association between the percentages spent by Japan and the United States.

$$H_0 : \rho = 0$$

- **Alternative Hypothesis (H_1):** There is a positive association between the percentages spent by Japan and the United States (one-tailed test).

$$H_1 : \rho > 0$$

Level of Significance: $\alpha = 5\%$ ($\alpha = 0.05$)

4 R Implementation & Calculation

To evaluate this computationally, we enter the dataset into R and utilize the `cor.test` function. We specify `method = "spearman"` and set `alternative = "greater"` to align with our alternative hypothesis of a positive relationship.

R Code:

```
# Input the percentage data for both countries
Japan <- c(0.8, 1.2, 0.7, 3.8, 0.4, 2.9, 1.9, 1.9, 1.6, 2.7, 5.1, 3.0, 2.6, 4.5)
USA <- c(0.4, 0.5, 1.3, 4.7, 0.7, 2.2, 0.5, 1.4, 1.3, 5.8, 4.8, 3.2, 1.2, 9.0)

# Perform Spearman's rank correlation test (one-tailed: positive association)
cor.test(Japan, USA, method = "spearman", alternative = "greater", exact = FALSE)
```

5 Result & Output

Executing the R code yields the following output:

Result Output:

Spearman's rank correlation rho

```
data: Japan and USA
S = 89.924, p-value = 0.0002663
alternative hypothesis: true rho is greater than 0
sample estimates:
      rho
0.8037491
```

(Note: I corrected the "not equal to 0" typo from your handwritten output to "greater than 0" so it matches your H_1 hypothesis and the "greater" argument in your R code).

6 Interpretation & Conclusion

- **Test Statistics:** The value of the test statistic is $S = 89.924$, and the sample estimate for Spearman's rho (ρ) is approximately 0.8037, indicating a strong positive correlation.
- **Decision Rule:** We evaluate the p-value against our assumed level of significance ($\alpha = 0.05$).
- **Conclusion:** Since the p-value (0.0002663) is strictly less than the level of significance ($\alpha = 0.05$), we reject the null hypothesis (H_0). We conclude at the 5% level of significance that there is a significant positive relationship between the R&D percentages spent by Japan and the United States across these industries.

PRACTICAL-8: Friedman's Two-Way ANOVA by Ranks

1 Aim

To test the null hypothesis that the effect of the group size is the same in terms of the total number of questions asked using the Friedman Test.

2 Problem Statement (Experiment)

An important factor in raising small children is to develop their ability to ask questions, especially in groups so that they will have this skill when they start school. A study of group size and the number of questions asked by preprimary children in a classroom atmosphere with a familiar person as a teacher consists of dividing 46 children randomly into four mutually exclusive groups of sizes 24, 12, 6, and 4.

The total number of questions asked by all members of each group is recorded for 30 minutes on each of eight different days. For the data shown in the table below, test the null hypothesis that the effect of the group size is the same in terms of the total number of questions asked.

Given Data:

Day	Group Size 24	Group Size 12	Group Size 6	Group Size 4
1	14	23	26	30
2	19	25	25	33
3	12	22	29	28
4	17	21	28	22
5	16	24	28	32
6	18	23	27	36
7	18	26	27	26
8	16	22	30	32

3 Theory & Mathematical Foundation

3.1 Introduction to the Test

Friedman's two-way ANOVA by ranks is a non-parametric alternative to the repeated measures (or randomized block) ANOVA. It is appropriate for testing the null hypothesis of equal treatment effects for data of n treatments applied across k blocks (or days, in this case).

3.2 Hypotheses

- **Null Hypothesis (H_0):** There is no difference in treatment effects across the groups.

$$H_0 : \theta_1 = \theta_2 = \cdots = \theta_n$$

- **Alternative Hypothesis (H_1):** At least two treatment effects differ significantly.

3.3 Test Statistic (S and Q)

For n treatments and k blocks, the data within each block is ranked from 1 to n . Let R_j be the sum of the ranks for the j -th treatment. The uncorrected sum of squared deviations (S) is given as:

$$S = \sum_{j=1}^n R_j^2 - \frac{k^2 n(n+1)^2}{4}$$

If ties are present to a significant extent, we use mid-ranks. The test statistic (Q) that incorporates the correction for ties is:

$$Q = \frac{12(n-1)S}{kn(n^2-1) - \sum \sum t(t^2-1)}$$

Where the double sum is extended over all sets of t tied ranks in each of the k blocks. Under H_0 , this statistic approximately follows a Chi-square (χ^2) distribution with $n-1$ degrees of freedom.

3.4 Multiple Comparisons (Post-Hoc)

If the null hypothesis is rejected, the procedure to declare that treatments i and j are significantly different in effect is if the absolute difference in their rank sums exceeds a critical value:

$$|R_i - R_j| \geq z \sqrt{\frac{kn(n+1)}{6}}$$

4 R Implementation & Calculation

To perform this in R, the data is structured into a matrix where rows represent blocks (Days) and columns represent treatments (Group Sizes).

R Code:

```
# Input the data into a matrix (8 rows, 4 columns) filled row by row
data <- matrix(c(14, 23, 26, 30,
                19, 25, 25, 33,
                12, 22, 29, 28,
                17, 21, 28, 22,
                16, 24, 28, 32,
                18, 23, 27, 36,
                18, 26, 27, 26,
                16, 22, 30, 32),
              nrow = 8, byrow = TRUE)

# Perform the Friedman Rank Sum Test
friedman.test(data)
```

5 Result & Output

Executing the R code yields the following output:

Result Output:

Friedman rank sum test

data: data

Friedman chi-squared = 20.577, df = 3, p-value = 0.0001289

6 Interpretation & Conclusion

- **Test Statistics:** We get $\chi^2 = 20.577$ with 3 degrees of freedom ($n - 1 = 4 - 1 = 3$).
- **Decision Rule:** The calculated p-value is 0.0001289. Since $p < 0.05$ (and even $p < 0.001$), it falls into the critical rejection region.
- **Conclusion:** We reject the null hypothesis at the 5% level of significance. There is a statistically significant difference in the number of questions asked based on group size. By observing the raw data and resulting ranks, it appears that the larger the group size, the fewer the questions asked.

PRACTICAL-9: Kendall's Coefficient of Concordance & Spearman's Rank Correlation

1 Aim

To find the Kendall's Coefficient of Concordance (W) and the Spearman coefficient of rank correlation (ρ).

2 Problem Statement (Experiment)

Eight graduate students are each given examinations in quantitative reasoning, vocabulary, and reading comprehension. Their scores are listed below. It is frequently claimed that a person who excels in quantitative reasoning is not as capable with verbal abilities, and vice versa, yet a truly intelligent person must possess all of these abilities.

- **Test 1:** Test these data to see if there is an association between all three scores.
- **Test 2:** Does there seem to be an indirect (negative) relationship between quantitative and verbal abilities?

Given Data:

Student	1	2	3	4	5	6	7	8
Quantitative	90	60	45	48	58	72	25	85
Vocabulary	62	81	92	76	70	75	95	72
Reading	60	91	85	81	60	76	93	80

3 Theory & Mathematical Foundation

3.1 Kendall's Coefficient of Concordance (W)

Kendall's Coefficient of Concordance (W) provides a measure of agreement between several samples, raters, or rankings. It determines the degree of dependence among the variables. It ranges between 0 and 1 and is a linear function of the average coefficient of rank correlation for all possible pairs of rankings.

Formula:

$$W = \frac{12S}{k^2n(n^2 - 1)}$$

Where:

- k = number of sets of rankings (number of variables/raters)
- n = number of entities (objects or individuals)
- $S = \sum \left[R_j - \frac{k(n+1)}{2} \right]^2$
- R_j = Sum of ranks given to the j -th object across all k raters.

Test of Significance based on W : For moderately large samples, the test statistic follows a Chi-square (χ^2) distribution:

$$Q = k(n - 1)W \sim \chi_{n-1}^2(\alpha)$$

3.2 Hypotheses (For Kendall's W)

- **Null Hypothesis (H_0):** There is no association (agreement) between the scores.
- **Alternative Hypothesis (H_1):** There is an association between the scores.

4 R Implementation & Calculation

To evaluate the overall association, we calculate Kendall's W . To evaluate the specific claim regarding quantitative vs. verbal abilities, we create a new combined **verbalscore** (Vocabulary + Reading) and calculate Spearman's rank correlation against the quantitative score.

R Code:

```
# Load the necessary library for Kendall's W
library(irr)

# Input the dataset
data1 <- data.frame(
  Quantitative = c(90, 60, 45, 48, 58, 72, 25, 85),
  Vocabulary = c(62, 81, 92, 76, 70, 75, 95, 72),
  Reading = c(60, 91, 85, 81, 60, 76, 93, 80)
)

# 1. Test overall association using Kendall's Coefficient of Concordance
kendall(data1)

# 2. Test indirect relationship between Quantitative and Verbal abilities
verbalscore <- data1$Vocabulary + data1$Reading
cor.test(data1$Quantitative, verbalscore, method = "spearman")
```

5 Result & Output

Executing the code in R yields the following two outputs:

Output 1: Kendall's Coefficient of Concordance W

```
Subjects = 8
Raters = 3
W = 0.138
Chisq(7) = 2.89
p-value = 0.895
```

Output 2: Spearman's Rank Correlation ρ

Spearman's rank correlation rho

data: data1\$Quantitative and verbalscore

S = 158, p-value = 0.009242
alternative hypothesis: true rho is not equal to 0
sample estimates:
rho
-0.8809524

6 Interpretation & Conclusion

Part I: Overall Association (Kendall's W)

- **Test Statistic:** The coefficient of concordance is $W = 0.138$. This shows a very low agreement between the rankings across the three tests.
- **Significance:** The value of the test statistic is $\chi^2 = 2.89$ with 7 degrees of freedom. The resulting p-value is 0.895.
- **Decision:** Since the p-value (0.895) is strictly greater than the level of significance α (0.05), the null hypothesis (H_0) is accepted.
- **Conclusion:** It can be concluded that there is no overall simultaneous association among the three scores.

Part II: Quantitative vs. Verbal Abilities (Spearman's ρ)

- **Test Statistic:** Spearman's rank correlation coefficient has been used to find the indirect relationship between quantitative and combined verbal abilities. The sample estimate for the coefficient is $\rho = -0.8809524$.
- **Significance:** The calculated p-value is 0.009242, which is less than 0.05, making this result statistically significant.
- **Conclusion:** The coefficient signifies a strong negative dependence (indirect relationship) between quantitative and verbal abilities, supporting the claim that individuals excelling in quantitative reasoning in this sample tend to have lower verbal scores, and vice versa.

PRACTICAL-10: Kendall's Coefficient of Concordance (Incomplete Rankings)

1 Aim

To calculate Kendall's Coefficient of Concordance (W) as a measure of agreement and to test the null hypothesis.

2 Problem Statement (Experiment)

A taste-test experiment to compare seven different kinds of wine is to be designed such that no taster will be asked to rank more than three different kinds. So we have $n = 7$ and $m = 3$. If each pair of wines is to be compared only once so that $\lambda = 1$, the required number of tasters is $k = \frac{\lambda n(n-1)}{m(m-1)} = 7$.

A balanced design was used and the rankings given are shown below. Calculate Kendall's Coefficient of Concordance as a measure of agreement between rankings and test the null hypothesis of no agreement.

Given Data:

Taster	Wine A	Wine B	Wine C	Wine D	Wine E	Wine F	Wine G
1	1	2	3				
2	1			3	2		
3	1					2	3
4		2			3	1	
5		1		3			2
6			2	1		3	
7			1		3		2
Total (R_j)	3	5	9	7	8	4	6

3 Theory & Mathematical Foundation

3.1 Introduction to the Test

Kendall's coefficient of concordance (W) provides a descriptive measure of agreement for incomplete sets of n rankings (often referred to as Durbin's test for Balanced Incomplete Block Designs). It is used when it is impractical for every judge to rank every single object.

3.2 Design Parameters

For a balanced incomplete block design, the parameters are defined as:

- n : Total number of objects presented for ranking (wines)
- m : Number of objects ranked by each judge
- k : Total number of judges (tasters)

- r : Number of times each object is ranked overall
- λ : Number of times each pair of objects is ranked together

3.3 Formula for Concordance (W)

The coefficient of concordance for incomplete rankings is calculated as:

$$W = \frac{12 \sum_{j=1}^n R_j^2 - 3r^2n(m+1)^2}{\lambda^2n(n^2-1)}$$

Where R_j is the sum of the ranks given to the j^{th} object.

3.4 Hypotheses & Test Statistic

- **Null Hypothesis (H_0):** There is no agreement between the rankings.
- **Alternative Hypothesis (H_1):** There is agreement between the rankings.

To test the significance of W , we use a Chi-square (χ^2) test statistic, denoted as Q :

$$Q = \frac{\lambda(n^2-1)W}{m+1} \sim \chi_{n-1}^2$$

Under the null hypothesis, Q follows a Chi-square distribution with $n-1$ degrees of freedom.

4 R Implementation & Calculation

To computationally evaluate this, we apply the mathematical formulas directly using R.

R Code:

```
# Input the sum of ranks for each wine
R <- c(3, 5, 9, 7, 8, 4, 6)

# Define design parameters
n <- 7      # Total number of wines
m <- 3      # Number of wines tasted by each judge
r <- 3      # Number of times each wine was tasted
lam <- 1    # Lambda (pair appearances)

# Calculate Kendall's W for incomplete rankings
W <- (12 * sum(R^2) - 3 * r^2 * n * (m+1)^2) / (lam^2 * n * (n^2 - 1))

# Calculate Chi-square test statistic (Q)
chisq <- lam * (n^2 - 1) * W / (m + 1)

# Calculate the p-value
p <- pchisq(chisq, df = n - 1, lower.tail = FALSE)

# Print results
cat("W =", W, "\n")
cat("Chi-square =", chisq, "\n")
cat("p-value =", p, "\n")
```

5 Result & Output

Executing the R code yields the following calculated values:

Result Output:

$W = 1$

Chi-square = 12

p-value = 0.0619688

6 Interpretation & Conclusion

- **Test Statistics:** We obtained a coefficient of $W = 1$, which describes a state of perfect agreement among the tasters given the constraints of the incomplete block design. The Chi-square test statistic is $Q = 12$ with 6 degrees of freedom ($n - 1 = 6$).
- **Significance & Decision:** The calculated p-value is 0.0619688. Because this p-value lies between $0.05 < p < 0.10$, the result is not statistically significant at the strict $\alpha = 0.05$ level, meaning we technically fail to reject the null hypothesis at the 5% level.
- **Conclusion:** However, the result is significant at the 10% level. Despite the perfect descriptive agreement ($W = 1$), the statistical power is limited by the small sample design. Therefore, there is strong indicative evidence of agreement between the rankings, even if it falls just shy of the traditional 5% significance threshold.